

B28: SQL SuperHero Program : Session 9

1 message

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Aggregate function:

What is Aggregate Functions in SQL?

SQL aggregation function is used to perform the calculations on columns of a table. After the calculation, It returns a single value.

SQL aggregation function is also used to summarize the data.

max, min, count, sum, avg

1. **Max** : Returns maximum value of expr

Example

The following query returns the max salary from the employees.

```
select max(sal) "Max Salary" from emp;
```

2. **Min**: Returns minimum value of expr.

Example

The following query returns the minimum salary from the employees.

```
select min(sal) "Min Salary" from emp;
```

3. Count: Whenever you heard, how many means count

4. SUM: total, sum

5. AVG : means Average

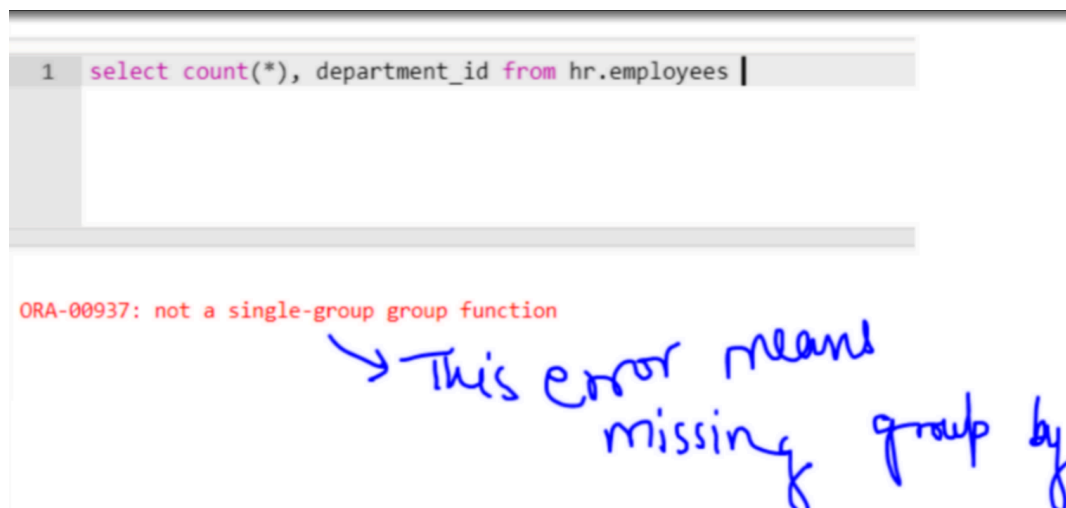
whenever you use aggregate function in SQL query and also fetching other columns than you must also used GROUP BY.

Always see the keyword Each/Every. Whenever listen these keywords means we have to group by on that column.

- **Each department - Group By department**
- **Each customer - Group by customer**
- **Every city - Group By city**
- **Every project - Group by project**
- **Every job id - Group By Job id**

we have to use group by with Aggregate function.

```
SELECT count(*), department_id FROM hr.employees;
```



-> **not a single-group group function**

Whenever this error comes, it means we are missing group by clause.

--> We are using an aggregate function and not using group by in query. It will give me "Not a single group-group function"

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Difference between count(1) and count(*)?

nothing, they are the same, incur the same amount of work -- do the same thing, take the same amount of resources.

```
1 create table test (id number, name varchar2(10));
2
3 insert into test values (null, 'A');
4 insert into test values (1, 'A');
5 insert into test values (2, 'B');
6 insert into test values (3, 'C');
7 insert into test values (4, null);
8
9 select count(*), count(1) from test
```

COUNT(*)	COUNT(1)
5	5

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*even for null values no difference.
Both are same*

```
select count(*), count(1), count(9), count(100), count(COMMISSION_PCT) from hr.employees
```

COUNT(*)	COUNT(1)	COUNT(9)	COUNT(100)	COUNT(COMMISSION_PCT)
107	107	107	107	35

See the above query. Count function used with any column then it will not consider any null records and gives not null values. Like count(commission_pct) = 35 because in table we have 35 records where commission_pct is not null

Practice Exercise:

<https://www.w3resource.com/sql-exercises/sql-aggregate-functions.php>